Individual Diagnostic Requirement Specification

CN200 ABS/ESP

Version: V1.4

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Requirement wording conventions

Within this document the following conventions are applied when describing specific compliance requirements:

The word “Should” denotes a preference or desired conformance.

The word “Shall” is used in the following ways:

1. To state a binding requirement on the special network, or the ECUs which comprise the network, which is verifiable by external manipulation and/or observation of an ECU or the network.

2. To state a binding requirement upon an ECU’s requirements document that is verifiable through a review of the document.

The word “Must” is used to state a binding requirement upon ECUs on the network which have a corresponding ECU requirements document. These requirements will be verified as part of the component verification.

NOTE: In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence.

NOTE: Nothing in this specification supersedes applicable laws and regulations unless a specific exemption has been obtained.

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History record of the document

The following table contains the revision history for this specification:

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Introduction

Scope

The detailed description about this chapter can be found in chapter 1.1 of the document *CN200 General Diagnostic Requirement Specification*.

NOTE: For CN200 vehicle, ABS needn't to realize the software download function, but ABS shall comply with the requirements for the non-programmable ECU, which is defined in chapter 16 of the document *CN200 General Diagnostic Requirement Specification*.

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Normative references

This document references ISO, GM, SGMW and other publications as listed in the following sub sections.

ISO Publications

[ISO 11898-2]	Road Vehicles - Controller Area Networks (CAN) - Part 2: High-speed medium access unit ISO 2003
[ISO 15765-1]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information ISO 2004
[ISO 15765-2]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services ISO 2004
[ISO 15765-3]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of diagnostic services ISO 2004
[ISO 14229]	Road Vehicles - Unified diagnostic services (UDS) - Specification and requirements ISO 2006
[ISO 15031-5]	Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 5: Emission-related diagnostic services ISO 2005
[ISO 15031-6]	Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 6: Diagnostic trouble code definitions ISO 2005

GM Publications

GMW 3107	GMLAN Functional Handler Specification
GMW 3104	GMLAN Communication Strategy Specification
GMW 3110	GMLAN Enhanced Diagnostic Test Mode Specification
GMW 3122	GM High Speed CAN Physical Layer Specification

SGMW Standard

CN200 General Diagnostic Requirement Specification

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Terms, definitions and abbreviations

ISO Terms

This document makes use of terms defined in the ISO 15765-2, ISO 15765-3 and ISO 14229-1.

Specific Terms Defined By This Document

Table 0-1Specific Terms Defined By This Document

Abbreviation	Description
ABS	**A**nti-lock **B**rake **S**ystem
BCM	**B**ody **C**ontrol **M**odule
EPS	**E**lectronic **P**ower **S**teering
IC	**I**nstrument **C**luster
SDM	Restraint **S**ensing and **D**iagnostic **M**odule
TCM	**T**ransmission **C**ontrol **M**odule
ECM	**E**ngine **C**ontrol **M**odule

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General requirement

All the general requirements, with which the tester and ABS shall comply, can be found in section 4 of the document *CN200 General Diagnostic Requirement Specification*.
It is possible for ABS supplier to add the additional definitions for general requirement in this section. For CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

Diagnostic CAN identifiers definition

The following table provides an overview about the definition of ABS diagnostic CAN identifier.

Table 0-1 ABS diagnostic CAN identifier

ECU		Functionally addressed request from tester to ECUs	Physically addressed request from tester to ECU	Response from ECU to tester
Name	Emission-related OBDECU (YES/NO)			
ABS	NO	0x7DF	0x720	0x728

ECU Diagnostic Address definition

The following table provides an overview about the definition of ABS diagnostic Address.

Table 0-2 ABS Diagnostic Address

ECU Name	Description	Value	Comments
ABS	ECUDiagnosticAddress	0x29	ABS diagnostic address, which is defined in *CN200 General Diagnostic Requirement Specification*.

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Physical layer requirement

All the physical layer requirements, with which the tester and ABS shall comply, can be found in section 5 of the document *CN200 General Diagnostic Requirement Specification*.
It is possible for ABS supplier to add the additional definitions for physical layer requirement in this section. For CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Data link layer requirement

All the data link layer requirements, with which the tester and ABS shall comply, can be found in section 6 of the document *CN200 General Diagnostic Requirement Specification*.
It is possible for ABS supplier to add the additional definitions for data link layer requirement in this section. For CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Network layer requirement

All the network layer requirements, with which the tester and ABS shall comply, can be found in section 7 of the document *CN200 General Diagnostic Requirement Specification*.
It is possible for ABS supplier to add the additional definitions for network layer requirement in this section. For CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Application and session layers requirement

All the application and session layer requirements, with which the tester and ABS shall comply, can be found in section 8 of the document *CN200 General Diagnostic Requirement Specification*.
It is possible for ABS supplier to add the additional definitions for application and session layer requirement in this section. For CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

Diagnostic Service Table Overview

The table followed (Table 0-1) shows the services which are allowed during the defaultSession, programmingSession and extendedSession of ABS. ABS is allowed to support more other diagnostic services per session after confirmed by SGMW.

Table 0-1 Diagnostic Service Table Overview

Service Name	SID (Hex)	Sub-function (Hex)	Session			Security Access[1]			Addressing method	
			Default	Programming	Extended	Level 1	Level 2	Level 3	Physical	Functional
Diagnostic and communication management functional unit										
DiagnosticSessionControl	10	01	Y[2]	-	Y	N[2]	N	N	Y	Y
		-	-	-	-	-	-	-	-	-
		03	Y	-	Y	N	N	N	Y	Y
ECUReset	11	01	Y	-	Y	N	N	N	Y	Y
		03	Y	-	Y	N	N	N	Y	Y
SecurityAccess	27	01/02	N	-	Y	N	N	N	Y	N
		-	-	-	-	-	-	-	-	-
		-	-	-	-	-	-	-	-	-
CommunicationControl	28	00/03	N	-	Y	N	N	N	Y	Y
TesterPresent	3E	00/80	Y	-	Y	N	N	N	Y	Y
AccessTimingParameter	83	-	-	-	-	-	-	-	-	-

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Service Name	SID (Hex)	Sub-function (Hex)	Session			Security Access[1]			Addressing method	
			Default	Programming	Extended	Level 1	Level 2	Level 3	Physical	Functional
SecuredDataTransmission	84	-	-	-	-	-	-	-	-	-
ControlDTCSetting	85	01/02	N	-	Y	N	N	N	Y	Y
ResponseOnEvent	86	-	-	-	-	-	-	-	-	-
LinkControl	87	-	-	-	-	-	-	-	-	-
ProgrammingMode	A5	01/03	N	-	Y	N	N	N	Y	Y
Data transmission functional unit										
ReadDataByIdentifier	22	-	Y	-	Y	N	N	N	Y	Y[3]
ReadMemoryByAddress	23	-	-	-	-	-	-	-	-	-
ReadScalingDataByIdentifier	24	-	-	-	-	-	-	-	-	-
ReadDataByPeriodicIdentifier	2A	-	-	-	-	-	-	-	-	-
DynamicallyDefineDataIdentifier	2C	-	-	-	-	-	-	-	-	-
WriteDataByIdentifier	2E	-	N	-	Y	Y[4]	N	N	Y	N
WriteMemoryByAddress	3D	-	-	-	-	-	-	-	-	-
Stored data transmission functional unit										
ClearDiagnosticInformation	14	-	Y	-	Y	N	N	N	Y	Y
ReadDTCInformation	19	01/02/04/06/0A	Y	-	Y	N	N	N	Y	Y
InputOutput control functional unit										
InputOutputControlByIdentifier	2F	-	-	-	-	-	-	-	-	-
Remote activation of routine functional unit										
RoutineControl	31	01/02/03	N	-	Y	N	N	N	Y	N
Upload download functional unit										

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Service Name	SID (Hex)	Sub-function (Hex)	Session			Security Access[1]			Addressing method	
			Default	Programming	Extended	Level 1	Level 2	Level 3	Physical	Functional
RequestDownload	34	-	-	-	-	-	-	-	-	-
RequestUpload	35	-	-	-	-	-	-	-	-	-
TransferData	36	-	-	-	-	-	-	-	-	-
RequestTransferExit	37	-	-	-	-	-	-	-	-	-

NOTE：

1. The definition of the security access level can be found in chapter 10.3.2.2 of the document *CN200 General Diagnostic Requirement Specification*.
2. “Y”：Support；“N”：Not Supported.
3. If the ReadDataByIdentifier (22 hex) service request message is a single frame, the functional addressing method shall be supported; otherwise, the functional addressing method is not required to be supported.
4. The WriteDataByIdentifier (2E hex) service in ABS shall need the security access, and the security level needed by this service depends on each DID, which is defined in Annex C.1 and Annex C.2 of this document.

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Diagnostic service documentation structure

All the requirements about diagnostic service documentation structure, with which the tester and ABS shall comply, can be found in section 9 of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions for diagnostic service documentation structure in this section. For CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Diagnostic and communication management functional unit

DiagnosticSessionControl (10 hex)

Request message

Table 0-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Request Service Id	10
#2	DiagnosticSessionType=[DefaultSession ExtendedDiagnosticSession]	 01 03

Positive response message

Table 0-2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Response Service Id	50
#2	DiagnosticSessionType=[DefaultSession ExtendedDiagnosticSession]	 01 03
 #3 #4 #5 #6	sessionParameterRecord[]#1=[$P2_{CAN_ECU_max}$(high byte) $P2_{CAN_ECU_max}$(low byte) $P2^*_{CAN_ECU_max}$(high byte) $P2^*_{CAN_ECU_max}$(low byte)]	 00 32 01 F4

NOTE: The definition about parameter $P2_{CAN_ECU_max}$ and $P2^*_{CAN_ECU_max}$ can be found in chapter 8.2.2 of the document *CN200 General Diagnostic Requirement Specification* and ISO 15765-3. These two parameters in positive response message are the ones that are used by ABS internally, if the value of this two parameters which is used by ABS are different from the definition in this document, it should be determined by SGMW and ABS Supplier together.

Supported negative response codes

Table 0-3 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

ECUReset (11 hex)

Request message

Table 0-4 Request message definition

A_Data Byte	Parameter name	Hex value
#1	ECUReset Request Service Id	11

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#2	resetType =[hardReset softReset]	 01 03

Positive response message

Table 0-5 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ECUReset Response Id	51
#2	resetType =[hardReset softReset]	 01 03

Supported negative response codes

Table 0-6 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ABS to implement the request message shall be defined by the ABS supplier.*

SecurityAccess (27 hex)

NOTE: This service, as well as MEC provides a protected mechanism to access the diagnostic services. The detailed definition about MEC can be found in chapter 16.1.3.2.5 of the document *CN200 General Diagnostic Requirement Specification*. The algorithm should be provided by SGMW.

Request message

Table 0-7 Request message definition

A_DataByte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType=requestSeed	01

Table 0-8 Request message definition

A_DataByte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType =sendKey	02
 #3 #4 #5 #6	securityKey[]=[key #1(MSB) key #2 key #3 key#4(LSB)]	 00-FF : : 00-FF

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Positive response message

Table 0-9 Positive response message definition (sub-function = requestSeed)

A_DataByte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType=requestSeed	01
#3 #4 #5 #6	securitySeed[]=[seed#1(MSB) seed#2 seed#3 seed#4(LSB)]	 00-FF : : 00-FF

Table 0-10 Positive response message definition (sub-function = sendKey)

A_DataByte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType =sendKey	02

Supported negative response codes

Table 0-11 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The correct condition for ABS to implement the request message shall be defined by the ABS supplier.*
24	**requestSequenceError**
35	**invalidKey**
36	**exceededNumberOfAttempts**
37	**requireTimeDelayNotExpired**

CommunicationControl (28 hex)

Request message

Table 0-12 Request message definition

A_DataByte	Parameter Name	Hex value
#1	CommunicationControl Request Service Id	28
#2	controlType=[enableRxAndTx disableRxAndTx]	 00 03
#3	communicationType=networkManagementCommunicationMessages and normalCommunicationMessages	03

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Positive response message

Table 0-13 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	CommunicationControl Response Service Id	68
#2	controlType=[enableRxAndTx disableRxAndTx]	 00 03

Supported negative response codes

Table 0-14 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
31	**requestOutOfRange**

TesterPresent (3E hex)

Request message

Table 0-15 Request message definition

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Request Service Id	3E
#2	zeroSubFunction=[zeroSubFunction (suppressPosRspMsgIndicationBit =0) zeroSubFunction (suppressPosRspMsgIndicationBit =1)]	 00 80

Positive response message

Table 0-16 Positive response message definition(sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =0))

A_DataByte	Parameter Name	Hex value
#1	TesterPresent Response Service Id	7E
#2	zeroSubFunction= zeroSubFunction (suppressPosRspMsgIndicationBit =0)	00

sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of suppressPosRspMsgIndicationBit =1.

Supported negative response codes

Table 0-17 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

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AccessTimingParameter (83 hex)

N/A

SecuredDataTransmission (84 hex)

N/A

ControlDTCSetting (85 hex)

Request message

Table 0-18 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ControlDTCSetting Request Service Id	85
#2	DTCSettingType=[On Off]	 01 02

Positive response message

Table 0-19 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ControlDTCSetting Response Service Id	C5
#2	DTCSettingType=[On Off]	 01 02

Supported negative response codes

Table 0-20 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

ResponseOnEvent (86 hex)

N/A

LinkControl (87 hex)

N/A

10.10 ProgrammingMode (A5 hex)

10.10.1 Request message

Table 0-21 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	ProgrammingMode Request Service Id	A5

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#2	programmingType = [verifyProgrammingConditions enableProgrammingMode]	 01 03

10.10.2 Positive response message

Table 0-22 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ProgrammingMode Positive Response Service Id	E5
#2	programmingType = [verifyProgrammingConditions enableProgrammingMode]	 01 03

10.10.3 Supported negative response codes

Table 0-23 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
22	**conditionsNotCorrect**
24	**requestSequenceError**

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Data transmission functional unit

ReadDataByIdentifier (22 hex)

NOTE: The maximum number of dataIdentifier that can be simultaneously requested is five.

NOTE: The DID which used to identify the general data record for all the ECUs called vehicle-manufacturer-specific DID, the definition about vehicle-manufacturer-specific DID can be found in chapter C.1 of Annex C. The DID which used to identify the individual data record for each ECU called ECU-supplier-specific DID, the definition about ECU-supplier-specific DID can be found in in chapter C.2 of Annex C.

Request message

Table 0-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Request Service Id	22
#2 #3	dataIdentifier[]#1=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
:	:	:
#n-1 #n	dataIdentifier[]#m[a]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
a. m≤5		

Positive response message

Table 0-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ReadDataByIdentifier Response Service Id	62
#2 #3	dataIdentifier[]#1=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4 : #(k-1)+4	dataRecord[]#1=[data #1 : data #k [a]]	00-FF : 00-FF
:	:	:
#n-(o-1)-2 #n-(o-1)-1	dataIdentifier[]#m[b]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#n-(o-1) : #n	dataRecord[]#m[b]=[data #1 : data #o [a]]	00-FF : 00-FF
a. The number of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C. b. m≤5		

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Supported negative response codes

Table 0-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ABS to implement the request message shall be defined by the ABS supplier.*

ReadMemoryByAddress (23 hex)

N/A

ReadScalingDataByIdentifier (24 hex)

N/A

ReadDataByPeriodicIdentifier (2A hex)

N/A

DynamicallyDefineDataIdentifier (2C hex)

N/A

WriteDataByIdentifier (2E hex)

Request message

Table 0-4 Request message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Request Service Id	2E
#2 #3	dataIdentifier[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF
#4 : #m+3	dataRecord[]=[data #1 : data #m[a]]	00-FF : 00-FF
a. The content of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C.		

Positive response message

Table 0-5 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	WriteDataByIdentifier Response Service Id	6E
#2 #3	dataIdentifier[]=[byte #1(MSB) byte #2(LSB)]	00-FF 00-FF

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Supported negative response codes

Table 0-6 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
33	**securityAccessDenied**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
72	**generalProgrammingFailure**

WriteMemoryByAddress (3D hex)

N/A

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Stored data transmission functional unit

ClearDiagnosticInformation (14 hex)

Request message

Table 0-1 Request message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Request Service Id	14
	groupOfDTC[]=[	
#2	groupOfDTCHighByte	FF
#3	groupOfDTCMiddleByte	FF
#4	groupOfDTCLowByte]	FF

Positive response message

Table 0-2 Positive response message definition

A_DataByte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Response Service Id	54

Supported negative response codes

Table 0-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** *The corresponding correct condition for ABS to implement the request message shall be defined by the ABS supplier.*

ReadDTCInformation (19 hex)

NOTE: ABS supplier should provide SGMW with DTC list and the document about DTC setting condition which ABS supported.

NOTE: The maximum number of DTC that ABS can store simultaneously is ten.

NOTE: If a DTC only supports a group snapshot data record, the maximum number of DTC snapshot record that ABS can store simultaneously should the same as the maximum number of DTC.

NOTE: From an attribute point of view, snapshot is divided into two parts: one is the general snapshot with one or more group data records for all the ECUs; Tthe other is local snapshot with one or more group data records for each specified ECU (ABS). The format and content about snapshot should be determined by SGMW and ABS Supplier together.

NOTE: If all fault memories are occupied, while the new DTC is generated, the Overflow strategy about stored DTCs and snapshot data records should be determined by SGMW and ABS supplier together.

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Request message

Table 0-4 Request message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=[reportNumberOfDTCByStatusMask reportDTCByStatusMask]	 01 02
#3	DTCStatusMask	00-FF

Table 0-5 Request message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
 #3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCSnapshotRecordNumber	FF[a]
a. A value of FF hex requests the ECU to report all stored DTCSnapshot data records at once.		

Table 0-6 Request message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
 #3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCExtendedDataRecordNumber	00-FF[a]
a. A value of FF hex requests the ECU to report all stored DTCExtendedData records in a single response message.		

Table 0-7 Request message definition (Sub-function = reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportSupportedDTC	0A

Positive response message

Table 0-8 Positive response message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportNumberOfDTCByStatusMask	01
#3	DTCStatusAvailabilityMask	FB
#4	ISO 15031-6 DTCFormat	00

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	DTCCount[]=[	
#5	DTCCountHighByte	00-FF
#6	DTCCountLowByte]	00-FF

Table 0-9 Positive response message definition (sub-function = reportDTCByStatusMask, reportSupportedDTC)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
	reportType=[	
#2	reportDTCByStatusMask	02
	reportSupportedDTC]	0A
#3	DTCStatusAvailabilityMask	FB
	DTCAndStatusRecord[] = [	
#4	DTCHighByte #1	00-FF
#5	DTCMiddleByte #1	00-FF
#6	DTCLowByte #1	00-FF
#7	statusOfDTC #1	00-FF
#8	DTCHighByte #2	00-FF
#9	DTCMiddleByte #2	00-FF
#10	DTCLowByte #2	00-FF
#11	statusOfDTC #2	00-FF
:	:	:
#n-3	DTCHighByte #m	00-FF
#n-2	DTCMiddleByte #m	00-FF
#n-1	DTCLowByte #m	00-FF
#n	statusOfDTC #m]	00-FF
NOTE: If reportType = reportSupportedDTC: 1. ABS should report the status of all the DTC's which is supported by ABS. 2. ABS should report the status at the time of receiving the requirement.		

Table 0-10 Positive response message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
	DTCAndStatusRecord[]=[	
#3	DTCHighByte	00-FF
#4	DTCMiddleByte	00-FF
#5	DTCLowByte	00-FF
#6	statusOfDTC]	00-FF
#7	DTCSnapshotRecordNumber#1	01
#8	DTCSnapshotRecordNumberOfIdentifiers#1	03-FF

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	DTCSnapshotRecord[]#1=[	
#9	ECU Voltage DataIdentifier(MSB)[a]	F0
#10	ECU Voltage DataIdentifier(LSB)[a]	11
#11	ECU Voltage[a]	00-FF
#12	Engine Speed DataIdentifier(MSB)[a]	F0
#13	Engine Speed DataIdentifier(LSB)[a]	12
#14	Engine Speed (High Byte)[a]	00-FF
#15	Engine Speed (Low Byte)[a]	00-FF
#16	Vehicle Speed DataIdentifier(MSB)[a]	F0
#17	Vehicle Speed DataIdentifier(LSB)[a]	13
#18	Vehicle Speed (High Byte)[a]	00-FF
#19	Vehicle Speed (Low Byte)[a]	00-FF
#20	dataIdentifier#4 byte #1 (MSB)	00-FF
#21	dataIdentifier#4 byte #2 (LSB)	00-FF
#22	snapshotData#4 byte #1	00-FF
:	:	:
#22+(p-1)	snapshotData#4 byte #p	00-FF
:	:	:
#r-2	dataIdentifier#m byte #1 (MSB)	00-FF
#r-1	dataIdentifier#m byte #2 (LSB)	00-FF
#r	snapshotData#m byte #1	00-FF
:	:	:
#r+(n-1)	snapshotData#m byte #n]	00-FF

a. This part is the general Snapshot for all ECUs, detailed definitions of general snapshot record can be found in chapter 12.2.3.2 of *CN200 General Diagnostic Requirement Specification*.

Table 0-11 Positive response message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
	DTCAndStatusRecord[]=[	
#3	DTCHighByte	00-FF
#4	DTCMiddleByte	00-FF
#5	DTCLowByte	00-FF
#6	statusOfDTC]	00-FF
#7	DTCExtendedDataRecordNumber #1[a]	01
#8	Fault occurrence counter[a]	00-FF
#9	DTCExtendedDataRecordNumber #2[a]	02
#10	Fault pending counter[a]	00-FF
#11	DTCExtendedDataRecordNumber #3[a]	03
#12	DTC Aging counter[a]	00-FF
#13	DTCExtendedDataRecordNumber #4[a]	04
#14	DTC Aged counter[a]	00-FF

a. This part is the DTCExtendedDataRecord for all ECUs, detailed definitions of DTC extended data record can be found in chapter 12.2.3.2 of *CN200 General Diagnostic Requirement Specification*. Whether the parameter appeared or not and the number of these parameters that appeared depend on the specific service requested.

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Supported negative response codes

Table 0-12 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**

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InputOutput control functional unit

InputOutputControlByIdentifier (2F hex)

N/A

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Remote activation of routine functional unit

RoutineControl (31 hex)

The service Routine Control is used by a diagnostic tool to start a routine, to stop a routine and to request the routine results.

NOTE: If the valves are activated for too long the ECU can be damaged. The maximum activation time of the valves must not exceed 30 seconds at an ambient temperature of 30°C. Using this service repeatedly can damage the valves or pump. Please keep this information in mind when using this service.

Request message

Table 0–1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	RoutineControl Request Service Id	31
#2	RoutineControlType=[startRoutine stopRoutine requestRoutineResults]	 01 02 03
 #3 #4	routineIdentifier[][a,b]=[byte#1(MSB) byte#2(LSB)]	 00-FF 00-FF
 #5 : #m+4	routineControlOptionRecord[][a,b,c]=[routineControlOption#1 : routineControlOption#m]	 00-FF : 00-FF

a. **If RoutineControlType = startRoutine:**
The parameter **routineControlOptionRecord** is **NOT** present if the routineIdentifier = 0x0402, 0x0405.
The detailed definition about **routineControlOptionRecord** can be found in chapter Annex F.2 of Annex F;

b. **If RoutineControlType = stopRoutine:**
RoutineControl for routineIdentifier=0x0402, 0x0403, 0x0404, 0x0405 can not be stopped by routineControlType = Stop Control.
The parameter **routineControlOptionRecord** is **NOT** present.

c. **If RoutineControlType = requestRoutineResults:**
the parameter **routineControlOptionRecord** is **NOT** present.

Positive response message

Table 0–2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	RoutineControl Response Service Id	71
#2	RoutineControlType=[startRoutine stopRoutine requestRoutineResults]	 01 02 03
 #3 #4	routineIdentifier[]=[byte#1 (MSB) byte#2 (LSB)]	 00-FF 00-FF

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#5 : #m+4	routineStatusRecord[][a]=[routineStatus #1 : routineStatus #m]	00-FF : 00-FF
a. The detailed definition about routineStatusRecord can be found in chapter F.3 of Annex F.		

Supported negative response codes

Table 0–3 Supported negative response codes

Hex	Description
7F	**serviceNotSupportedInActiveSession**
33	**securityAccessDenied**
12	**subFunctionNotSupported**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect** This code shall be returned if the criteria for the request InputOutputControl are not met. *The corresponding correct condition for ABS to implement the request message shall be defined by the ABS supplier.*

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Upload download functional unit

RequestDownload (34 hex)

N/A

RequestUpload (35 hex)

N/A

TransferData (36 hex)

N/A

RequestTransferExit (37 hex)

N/A

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ECU programming requirements and process

The detailed description about this chapter can be found in chapter 16 of the document *CN200 General Diagnostic Requirement Specification*.

NOTE: For CN200 vehicle, ABS needn't to realize the software download function, but ABS shall comply with the requirements for the non-programmable ECU, which is defined in chapter 16 of the document *CN200 General Diagnostic Requirement Specification*.

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Annex A (normative) Global parameter definitions

The detailed description about this Annex can be found in Annex A of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex A for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex B (normative) Diagnostic and communication management functional unit data parameter definitions

The detailed description about this Annex can be found in Annex B of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex B for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex C (normative) Data transmission functional unit data parameter definitions

C.1 Vehicle-manufacturer-specific DID definitions

The following table lists the detailed definition about vehicle-manufacturer-specific DID. The function and actual meaning about these DIDs can be found in the document of *CN200 General Diagnostic Requirement Specification*.

DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
		22	2E		
F016	ECUDiagnosticAddress	x	-	1	USN
F186	Active Diagnostic Session	x	-	1	USN
F18A	Supplier Identifier Information	x	-	7	ASCII
F18B	ECUManufacturingDate[a]	x	-	6	ASCII
F18C	ECUSerialNumber	x	-	20	ASCII
F190	VehicleIdentificationNumber	x	x^1	17	ASCII
F192	SupplierECUHardwareNumber	x	-	5	USN
F194	Supplier ECU Software Number	x	-	3	USN
F195	Supplier ECU Software Version Number	x	-	4	USN
F1A0	Manufacturers Enable Counter	x	x^1	1	USN
F1B2	Vehicle Manufacturing Date	x	x^1	4	BCD
F1B9	Subnet Configuration List on High Speed CAN	x	x^1	2	BIN
F1CB	EndModelPartNumber	x	-	4	USN

a. The definition about data length and data type for this DID is not able to follow the definition of the document *CN200 General Diagnostic Requirement Specification*, The definition in this document is only for ABS Supplier.

NOTE: "×": Support without security access; "x^n": Support with security access (A superscript number 'n' after 'x' indicates that the number of the security level); "-": Not Supported.

NOTE: For specific value of each DID, which should be written in SGMW's production line or supplier's production line can be found in Annex C.2 of the document *CN200 General Diagnostic Requirement Specification*.

C.1.1 DID 0xF016- ECUDiagnosticAddress

Byte No.	Description	Data Type	Value (Hex)
data #1	ECUDiagnosticAddress	USN	29

NOTE: ABS diagnostic address, which is defined in Annex C of the document *CN200 General Diagnostic Requirement Specification*.

C.1.2 DID 0xF186- Active Diagnostic Session

Byte No.	Description	Data Type	Value (Hex)
data #1	Active Diagnostic Session	USN	01=defaultSession 03=extendedSession

NOTE: The default value is 0x01 (default session).

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C.1.3 DID 0xF18A- Supplier Identifier Information

Byte No.	Description	Data Type	Value (Hex)
data #1	Supplier Identifier Information(high byte)	ASCII	the real number
data #2	Supplier Identifier Information	ASCII	the real number
data #3	Supplier Identifier Information	ASCII	the real number
data #4	Supplier Identifier Information	ASCII	the real number
data #5	Supplier Identifier Information	ASCII	the real number
data #6	Supplier Identifier Information	ASCII	the real number
data #7	Supplier Identifier Information(low byte)	ASCII	the real number
NOTE: This DID indicate the real number of supplier identifier information which used in SGMW.			

C.1.4 DID 0xF18B- ECUManufacturingDate

Byte No.	Description	Data Type	Value (Hex)
data #1	Year(high byte)	ASCII	the real year
data #2	Year(low byte)	ASCII	
data #3	Month(high byte)	ASCII	the real month
data #4	Month(low byte)	ASCII	
data #5	Day(high byte)	ASCII	the real day
data #6	Day(low byte)	ASCII	

For example, if ECUManufacturingDate is 2013.08.28. See the following.

Default Value	The real ECU Manufacturing Date						
Position	1	2	3	4	5	6	
ASCII	1	3	0	8	2	8	2013 = Year; 08 = Month; 28 = Day
Hex value	31	33	30	38	32	38	

a. The definition about data length and data type for this DID is not able to follow the definition of the document *CN200 General Diagnostic Requirement Specification*, The definition in this document is only for ABS Supplier.

NOTE: This DID indicate the real ECU manufacturing date in supplier's production line.

C.1.5 DID 0xF18C- ECU Serial Number

Byte No.	Description	Data Type	Value (Hex)
data #1	ECU Serial Number (high byte)	ASCII	the real number
:	:		:
data #20	ECU Serial Number (low byte)		the real number
NOTE: If the byte(s) is (are) not used, the default value for this (these) byte(s) shall be determined by supplier and SGMW together.			

C.1.6 DID 0xF190- VehicleIdentificationNumber

Byte No.	Description	Data Type	Value (Hex)
data #1	VehicleIdentificationNumber (high byte)	ASCII	the real number
:	:	:	:
data #17	VehicleIdentificationNumber (low byte)	ASCII	the real number
NOTE: The default value is 0x20 in ASCII.			

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C.1.7 DID 0xF192- Supplier ECU Hardware Number

Byte No.	Description	Data Type	Value (Hex)
data #1	Supplier ECU Hardware Number (high byte)	USN	the real number
:	:		:
data #5	Supplier ECU Hardware Number (low byte)		the real number
NOTE: If the byte(s) is (are) not used, the default value for this (these) byte(s) shall be determined by supplier and SGMW together.			

C.1.8 DID 0xF194- Supplier ECU Software Number

Byte No.	Description	Data Type	Value (Hex)
data #1	Supplier ECU Software Number (high byte)	USN	the real number
:	:		:
data #3	Supplier ECU Software Number (low byte)		the real number
NOTE: If the byte(s) is (are) not used, the default value for this (these) byte(s) shall be determined by supplier and SGMW together.			

C.1.9 DID 0xF195- Supplier ECU Software Version Number

Byte No.	Description	Data Type	Value (Hex)
data #1	Supplier ECU Software Version Number (high byte)	USN	the real number
:	:		:
data #4	Supplier ECU Software Version Number (low byte)		the real number
NOTE: If the byte(s) is (are) not used, the default value for this (these) byte(s) shall be determined by supplier and SGMW together.			

C.1.10 DID 0xF1A0- Manufacturers Enable Counter

Byte No.	Description	Data Type	Value (Hex)
data #1	Manufacturers Enable Counter	USN	0xFE~0x01: ABS's security system not armed; 0x00: ABS's security system armed.
NOTE: 1. The default value is 0xFE for Manufacturer production line. 2. The default value is 0x00 for After-sales dept.			

C.1.11 DID 0xF1B2- Vehicle Manufacturing Date

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Byte No.	Description	Data Type	Value (Hex)
data #1	Year(high byte)	BCD	the real year
data #2	Year(low byte)	BCD	
data #3	Month	BCD	the real month
data #4	Day	BCD	the real day
NOTE: The default value is 0x00000000.			

C.1.12 DID 0xF1B9- Subnet Configuration List on High Speed CAN

Byte No.	Description	Data Type	Value (Hex)	
data #1	Subnet Configuration List on High Speed CAN (high byte)	BIN	bit7	ECM node
			bit6	TCM node
			bit5	Reserved
			bit4	ABS node
			bit3	Reserved
			bit2	ESP node
			bit1	EPS node
			bit0	Reserved
data #2	Subnet Configuration List on High Speed CAN (low byte)	BIN	bit7	SDM node
			bit6	Reserved
			bit5	IC node
			bit4	ICE node
			bit3	BCM node
			bit2	Reserved
			bit1	Reserved
			bit0	Reserved
NOTE: 1. The default value is 0x0000 for ABS. 2. Each bit represents one ECU that can exist on the high speed CAN subnet. A logic '1' indicates that a given ECU is present on the vehicle. The reserved bit(s) shall be written as '0'.				

C.1.13 DID 0xF1CB- EndModelPartNumber

Byte No.	Description	Data Type	Value (Hex)
data #1	EndModelPartNumber	USN	the real number
:	:	:	:
data #4	EndModelPartNumber	USN	the real number
NOTE: This DID is used to identify the part number that represents the combination of ardware/software/calibrations present in the ECU as it is received in SGMW vehicle assembly plant.			

C.2 ECU-supplier-specific DID definitions

The following table lists the definition about ECU-supplier-specific DID. According to the requirements of the document *CN200 General Diagnostic Requirement Specification*, the range which ABS used is 0x2001~0x2FFF.

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DID (Hex)	Data Description	Supported by Service (Hex)		Data Length (Byte)	Data Type
		22	2E		
2D00	SpeedInformation	x	-	10	Hex
2D01	InputData	x	-	3	Hex
2D02	ActuationState	x	-	4	Hex&BIN
2D03	FillingInStatus	x	x^1	1	Hex
2D04	EOLStatus	x	x^1	1	Hex
2D05	SystemSensor (ESP only)	x	-	10	Hex
2D06	Variant Code (ESP only)	x	x^1	1	Hex
2D07	Speed Limit Disable/Enable	x	-	1	Hex
2D08	Disable Valve Relay Shutdown	x	-	1	Hex
2D10	SystemName	x	-	3	ASCII
NOTE: "×": Support without security access; "$\times^{n}$": Support with security access (A superscript number 'n' after 'x' indicates that the number of the security level); "-": Not Supported.					

C.2.1 DID 0x2D00- SpeedInformation

Byte No.	Description	Unit	Scaling	HEX limits	Physical limits
data #1	Vehicle speed (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #2	Vehicle speed (low Byte)				
data #3	Wheel speed FL (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #4	Wheel speed FL (low Byte)				
data #5	Wheel speed FR (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #6	Wheel speed FR (low Byte)				
data #7	Wheel speed RL (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #8	Wheel speed RL (low Byte)				
data #9	Wheel speed RR (high Byte)	km/h	0.05625	0x0000~0x120E	0~260
data #10	Wheel speed RR (low Byte)				
NOTE: If there is failure at any of the wheel speed sensor, the corresponding vehicle speed/ wheel speed will be invalid and the output will be 0x7FFF.					

C.2.2 DID 0x2D01- InputData

Byte No.	Description	Unit	Scaling	HEX limits	Physical limits
data #1	Battery Voltage	V	0.08	0x00~0xFF	0~20.4
Byte No.	**Description**	**Unit**	**Scaling**	**Comments**	
data #2	Brake Light Switch	-	-	0x00: OFF 0x01: ON	
data #3	ESP PATA Switch **(ESP only)**	-	-	0x00: OFF 0x01: ON	

C.2.3 DID 0x2D02- ActuationState

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Byte No.	Description	Data Type	Comments		
data #1	Valve Relay Status	Hex	0x00: OFF 0x01: ON		
data #2	Pump Motor Status	Hex	0x00: OFF 0x01: ON		
data #3	Valve Actuation State(high Byte)	BIN	bit7	Outlet Valve Rear Right (AVRR)	0 = inactive 1 = active
			bit6	Inlet Valve Rear Right (EVRR)	0 = inactive 1 = active
			bit5	Outlet Valve Rear Left(AVRL)	0 = inactive 1 = active
			bit4	Inlet Valve Rear Left(EVRL)	0 = inactive 1 = active
			bit3	Outlet Valve Front Right (AVFR)	0 = inactive 1 = active
			bit2	Inlet Valve Front Rear (EVFR)	0 = inactive 1 = active
			bit1	Outlet Valve Front Left(AVFL)	0 = inactive 1 = active
			bit0	Inlet Valve Front Left(EVFL)	0 = inactive 1 = active
data #4	Valve Actuation State(low Byte)	BIN	bit7	Unused	0
			bit6	Unused	0
			bit5	Unused	0
			bit4	Unused	0
			bit3	Valve HSV2 (ESP Only)	0 = inactive 1 = active
			bit2	Valve HSV1 (ESP Only)	0 = inactive 1 = active
			bit1	Valve USV2 (ESP Only)	0 = inactive 1 = active
			bit0	Valve USV1 (ESP Only)	0 = inactive 1 = active
NOTE: If the bit(s) is (are) not used, the default value is "0".					

C.2.4 DID 0x2D03- FillingInStatus

Byte No.	Description	Data Type	Default Value	Comments
data #1	FillingInStatus	Hex	0xFF	0x00 : Filling-in not completed 0xAA : Filling-in completed and ok 0xEE : Filling-in completed and not ok
NOTE: The data byte of delivery state from the supplier is 0xFF.				

C.2.5 DID 0x2D04- EOLStatus

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Byte No.	Description	Data Type	Default Value	Comments
data #1	EOLStatus	Hex	0xFF	0x00 : End-of-line test not completed 0xAA : End-of-line test completed and ok 0xEE : End-of-line test completed and not ok
NOTE: The data byte of delivery state from the supplier is 0xFF.				

C.2.6 DID 0x2D05- SystemSensor (ESP only)

Byte No.	Description	Unit	Scaling	Offset	Physical limits
data #1	Master Cyclinder Pressure(high Byte)	bar	0.0153	0	-42.5~425
data #2	Master Cyclinder Pressure(low Byte)				
data #3	Steering Angle(high Byte)	deg	0.1	0	-1440~1440
data #4	Steering Angle(low Byte)				
data #5	Yaw Rate(high Byte)	rad/s	0.00213	0	-2.1~2.1
data #6	Yaw Rate(low Byte)				
data #7	Lateral Acceleration(high Byte)	m/s^2	0.02712	0	-21.6~21.6
data #8	Lateral Acceleration(low Byte)				
data #9	Longitudinal Acceleration(high Byte)	m/s^2	0.02712	0	-21.6~21.6
data #10	Longitudinal Acceleration(low Byte)				

C.2.7 DID 0x2D06- Variant Code (ESP only)

Byte No.	Description	Data Type	Default Value	Comments
data #1	Variant Code	Hex	-	0x01 :*defined by SGMW and ABS Supplier together* 0x02 : *defined by SGMW and ABS Supplier together* 0x03 : *defined by SGMW and ABS Supplier together* 0x04: *defined by SGMW and ABS Supplier together* 0x05: *defined by SGMW and ABS Supplier together* 0x06: *defined by SGMW and ABS Supplier together*

C.2.8 DID 0x2D07- Speed Limit Disable/Enable

Byte No.	Description	Data Type	Default Value	Comments
data #1	Speed Limit Disable/Enable	Hex	-	0x00: Speed Limit Disabled 0x01: Speed Limit Enabled

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C.2.9 DID 0x2D08- Disable Valve Relay Shutdown

Byte No.	Description	Data Type	Default Value	Comments
data #1	Disable Valve Relay Shutdown	Hex	-	0x00: Valve Relay Shutdown Disabled 0x01: Valve Relay Shutdown Enabled

C.2.10 DID 0x2D10- SystemName

Byte No.	Description	Data Type	Default Value	Comments			
data #1	SystemName	ASCII	*To be defined by ABS Supplier*	ABS	0x41	ESP	0x45
data #2					0x42		0x53
data #3					0x53		0x50
NOTE: The default value for 0x2D10 should be written into ECU EEPROM by ABS Supplier.							

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Annex D (normative) Stored data transmission functional unit data parameter definitions

The detailed description about this Annex can be found in Annex D of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex D for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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Annex E (normative) Input output control functional unit data parameter definitions

E.1 InputOutputControlParameter definitions

The detailed description about this Annex can be found in Annex E.1 of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex E.1 for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

E.2 InputOutputControlByIdentifier (2F hex) controlOptionRecord definitions

N/A

E.3 InputOutputControlByIdentifier (2F hex) controlEnableMaskRecord definitions

N/A

E.4 InputOutputControlByIdentifier (2F hex) controlStatusRecord definitions

N/A

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Annex F (normative) Remote activation of routine functional unit data parameter definitions

F.1 RoutineIdentifier definition

The detailed description about this Annex can be found in Annex F.1 of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex F.1 for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

F.2 RoutineControl (31 hex) routineControlOption definitions

The following table lists the definition about routine identifier, According to the requirements of the document *CN200 General Diagnostic Requirement Specification*, the range which ABS used is 0x0200~0xDFFF.

RID (Hex)	Description	Supported routineControlType (Hex)		
		01	02	03
0300	DelayedActuatorControl	x	x	x
0301	EvacAndFill	x	x	x
0302	RepairBleed	x	x	x
0303	ActuatorTest	x	x	x
0304	DynamicTest	x	x	x
0305	WheelSpeedSensorsTest	x	x	x
0402	InertialSensorCalibration (ESP only)	x	-	x
0403	Speed Limit Disable/Enable	x	-	x
0404	Disable Valve Relay Shutdown	x	-	x
0405	SAS Calibration (ESP only)(Optional for ESP)	x	-	x
NOTE: "×": Support without security access; "×^{n}": Support with security access (A superscript number 'n' after 'x' indicates that the number of the security level); "-": Not Supported.				

The following sections (F.2.1~F.2.10) list the definition about routine control option.

F.2.1 RID 0x0300- DelayedActuatorControl

Byte No.	routineControlOptionRecord	Data Type	value
data #1	ValveStates(State 1)	BIN	bit7: Outlet Valve Rear Right bit6: Inlet Valve Rear Right bit5: Outlet Valve Rear Left bit4: Inlet Valve Rear Left bit3: Outlet Valve Front Right bit2: Inlet Valve Front Right bit1: Outlet Valve Front Left bit0: Inlet Valve Front Left **NOTE:** For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.

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Byte No.	routineControlOptionRecord	Data Type	value
data #2	DriverStates(State 1)	BIN	bit7: Unused bit6: Pump bit5: Unused bit4: Unused bit3: HSV2 (ESP only) bit2: HSV1 (ESP only) bit1: USV2 (ESP only) bit0: USV1 (ESP only) **NOTE:** For each control the corresponding bit has to be set to 1 to switch the actuator on and to 0 to switch the actuator off.
data #3	Activation time	DEC	Resolution: 20ms/bit, offset: 0.
data #4	ValveStates(State 2)	BIN	Same as data#1
data #5	DriverStates(State 2)	BIN	Same as data#2

NOTE: This request is used to control multiple drivers simultaneously. For each control two states are defined: state 1 and state 2. When the request was received the selected controls are set in state 1. After the activation time they are set to state 2. In comparison to the Actuator Test this service allows the tester to actuate one or more valves for a specified time. This feature allows the tester to run the customer specified End Of Line routine sequence instead of Pre-defined Periodic Actuator Control (EvacAndFill, RepairBleed) sequence.
NOTE: A "Failure" response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons.
NOTE: A control that has been started by a delay actuator control request can be stopped by a request with the ControlOption set to "Stop control" (Emergency Stop, 31h 02h F0h 00h).
NOTE: At each end of this service the tester has to send the following request to turn off all controls: 31h 01h F0h 00h 00h 00h 00h 00h 00h.

F.2.2 RID 0x0301- EvacAndFill

Byte No.	routineControlOptionRecord	Unit	Scaling	Offset	value
data #1	Evac and Fill time	S	1s/bit	0	2~150
data #2	Time until start of pump motor	S	1s/bit	0	2~150
	NOTE: 0xFF= no pump motor				

NOTE: This request is used to fill and drain the hydraulic unit.
NOTE: Time has to be between 2s and 150s. If Time exceeds this limit, the ECU uses the maximum defined value. After evac and filling time, the pump motor will run for 2 additional seconds, if the delay time in byte 5 is not 0xFF.

F.2.3 RID 0x0302- RepairBleed

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Repair Bleeding phase	DEC	0x01: phase1 0x02: phase2 0x03: phase3 0x04: phase4
data #2	Repetition counter	DEC	0x03: 3 loops 0x04: 4 loops 0x05: 5 loops

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Byte No.	routineControlOptionRecord	Data Type	value
NOTE: For parameter "Repair Bleeding phase", the repair bleeding function should always be executed in increasing phase sequence starting with Phase 1. A repetition of the same phase is allowed, but no phase shall be skipped. If this is not ensured, the pump might be damaged due to dry run. **NOTE:** For parameter "Repetition counter", this parameter defines the number of inner loop repetitions in phase 3 and 4. This parameter is always required, but only valid for Phase 3 and 4, for other phases, it is ignored.			

F.2.4 RID 0x0303- ActuatorTest

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Actuator 1 Value(high byte)	DEC	0x0000=OFF 0xFFFF=ON
data #2	Actuator 1 Value(low byte)	DEC	
data #3	Actuator 1 Identifier(high byte)	DEC	0x0000 = No actuation 0x0022 = Pump motor relay 0x0030 = Inlet valve front left 0x0032 = Outlet valve front left 0x0034 = Inlet valve front right 0x0036 = Outlet valve front right 0x0038 = Inlet valve rear right 0x003A = Outlet valve rear right 0x003C = Inlet valve rear left 0x003E = Outlet valve rear left 0x004E= HSV2 (ESP only) 0x0050= HSV1 (ESP only) 0x0052= USV2 (ESP only) 0x0054= USV1 (ESP only) 0x0002 = ABS Safety Lamp 0x0004 = ESP Safety Lamp (ESP Only) 0x000E = EBD Safety Lamp
data #4	Actuator 1 Identifier(low byte)	DEC	
data #5	Actuator 2 Value(high byte)	DEC	Same as data#1 and data#2
data #6	Actuator 2 Value(low byte)	DEC	
data #7	Actuator 2 Identifier(high byte)	DEC	Same as data#3 and data#4
data #8	Actuator 2 Identifier(low byte)	DEC	
data #9	Activation time(high byte)	DEC	Resolution: 10ms/bit, offset: 0. range: 10ms~327s
data #10	Activation time(low byte)	DEC	
data #11	Actuator 3 Value(high byte)	DEC	Same as data#1 and data#2
data #12	Actuator 3 Value(low byte)	DEC	
data #13	Actuator 3 Identifier(high byte)	DEC	Same as data#3 and data#4
data #14	Actuator 3 Identifier(low byte)	DEC	
data #15	Actuator 4 Value(high byte)	DEC	Same as data#1 and data#2
data #16	Actuator 4 Value(low byte)	DEC	
data #17	Actuator 4 Identifier(high byte)	DEC	Same as data#3 and data#4
data #18	Actuator 4 Identifier(low byte)	DEC	

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Byte No.	routineControlOptionRecord	Data Type	value
NOTE: This request is used to start Actuator Test. **NOTE:** The pump motor relay must not be actuated for times shorter than 20ms. Shorter actuations could damage the motor relay due to very high current during the starting phase of the motor. Strongly recommends only using an ActuationValue of 0% and 100% **NOTE:** A negative response has one of the following reasons: At least one valve or one driver doesn't exist or the control is not possible at this time because of safety reasons. **NOTE:** If the valve relay is out of order, the routine will be aborted.			

F.2.5 RID 0x0304- DynamicTest

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Actuator 1 Value(high byte)	DEC	0x0000=OFF 0xFFFF=ON
data #2	Actuator 1 Value(low byte)	DEC	
data #3	Actuator 1 Identifier(high byte)	DEC	0x0000 = No actuation 0x0022 = Pump motor relay 0x0030 = Inlet valve front left 0x0032 = Outlet valve front left 0x0034 = Inlet valve front right 0x0036 = Outlet valve front right 0x0038 = Inlet valve rear right 0x003A = Outlet valve rear right 0x003C = Inlet valve rear left 0x003E = Outlet valve rear left 0x004E= HSV2 (ESP only) 0x0050= HSV1 (ESP only) 0x0052= USV2 (ESP only) 0x0054= USV1 (ESP only) 0x0002 = ABS Safety Lamp 0x0004 = ESP Safety Lamp (ESP Only) 0x000E = EBD Safety Lamp
data #4	Actuator 1 Identifier(low byte)	DEC	
data #5	Actuator 2 Value(high byte)	DEC	Same as data#1 and data#2
data #6	Actuator 2 Value(low byte)	DEC	
data #7	Actuator 2 Identifier(high byte)	DEC	Same as data#3 and data#4
data #8	Actuator 2 Identifier(low byte)	DEC	
data #9	Activation time(high byte)	DEC	Resolution: 10ms/bit, offset: 0. range: 10ms~327s
data #10	Activation time(low byte)	DEC	
data #11	Actuator 3 Value(high byte)	DEC	Same as data#1 and data#2
data #12	Actuator 3 Value(low byte)	DEC	
data #13	Actuator 3 Identifier(high byte)	DEC	Same as data#3 and data#4
data #14	Actuator 3 Identifier(low byte)	DEC	
data #15	Actuator 4 Value(high byte)	DEC	Same as data#1 and data#2
data #16	Actuator 4 Value(low byte)	DEC	
data #17	Actuator 4 Identifier(high byte)	DEC	Same as data#3 and data#4
data #18	Actuator 4 Identifier(low byte)	DEC	
data #19	Delay Time(high byte)	DEC	Resolution: 10ms/bit, offset: 0. range: 10ms~327s
data #20	Delay Time(low byte)	DEC	
NOTE: This request is used to start Dynamic Test.			

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F.2.6 RID 0x0305- WheelSpeedSensorsTest

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Execution time (high byte)	DEC	Resolution: 10ms/bit, offset: 0.
data #2	Execution time (low byte)	DEC	
NOTE: This request is used to start Wheel Speed Sensors Test. **NOTE:** Execution time can't be 0, or else NRC13 will be returned.			

F.2.7 RID 0x0402- InertialSensorCalibration (ESP only)

Byte No.	routineControlOptionRecord	Data Type	value
-	-	-	routineControlOptionRecord is NOT present for routineIdentifier=0x0402.
NOTE: This request is used to start inertial sensor calibration. **NOTE:** This Routine 0x0402 cannot be stopped by ControlOption= Stop Control.			

F.2.8 RID 0x0403- Speed Limit Disable/Enable

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Speed Limit Disable/Enable	DEC	0xFF = Enable Speed Limit 0x00 = Disable speed Limit
NOTE: This request is Disable/Enable the speed Limit (10kph for Extended Mode). **NOTE:** This Routine 0x0403 cannot be stopped by ControlOption= Stop Control.			

F.2.9 RID 0x0404- Disable Valve Relay Shutdown

Byte No.	routineControlOptionRecord	Data Type	value
data #1	Disable Valve Relay Shutdown	DEC	0xFF = Re-enable Valve relay shutdown 0x00 = Disable Valve Relay Shutdown
NOTE: This Routine 0x0404 cannot be stopped by ControlOption= Stop Control.			

F.2.10 RID 0x0405- SAS Calibration (ESP only) (Optional for ESP)

Byte No.	routineControlOptionRecord	Data Type	value
-	-	-	routineControlOptionRecord is NOT present for routineIdentifier=0x0405.
NOTE: This request is used to calibrate the SAS (Steering Angle Sensor) **NOTE:** This Routine 0x0405 cannot be stopped by ControlOption= Stop Control.			

F.3 RoutineControl (31 hex) routineStatus definitions

The following sections (F.3.1~F.3.10) list the definition about routine status.

F.3.1 RID 0x0300- DelayedActuatorControl

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Byte No.	Description	Data Type	value
data #1	Supported Control /Routine Status	DEC	If routineControlType = startRoutine, then 01 = start successful 02 = already started 03 = not possible to start Else if routineControlType = stopRoutine, then 01 = stop successful 02 = already stopped or not start 03 = routine stopped abnormal Else if routineControlType = requestRoutineResults, then 01 = finished 02 = still running 03 = stopped or not start 04 = routine stopped abnormal

F.3.2 RID 0x0301- EvacAndFill

Byte No.	Description	Data Type	value
data #1	Supported Control /Routine Status	DEC	Same as the Comments in the table of F.3.1

F.3.3 RID 0x0302- RepairBleed

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.1

F.3.4 RID 0x0303- ActuatorTest

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.1
data #2	Result, actuator 1(high byte)	DEC	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #3	Result, actuator 1(low byte)	DEC	
data #4	Result, actuator 2(high byte)	DEC	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #5	Result, actuator 2(low byte)	DEC	
data #6	Result, actuator 3(high byte)	DEC	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #7	Result, actuator 3(low byte)	DEC	

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Byte No.	Description	Data Type	value
data #8	Result, actuator 4(high byte)	DEC	*The description should be defined by the ABS supplier and confirmed by SGMW.*
data #9	Result, actuator 4(low byte)	DEC	
NOTE: These bytes from data #2 to data #9 are only for RoutineControlType = requestRoutineResults and Supported Control / Routine Status = "finished".			

F.3.5 RID 0x0304- DynamicTest

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.1
data #2	delta wheel speed FL (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #3	delta wheel speed FL (low byte)	DEC	
data #4	delta wheel speed FR (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #5	delta wheel speed FR (low byte)	DEC	
data #6	delta wheel speed RL (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #7	delta wheel speed RL (low byte)	DEC	
data #8	delta wheel speed RR (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #9	delta wheel speed RR (low byte)	DEC	
NOTE: These bytes from data #2 to data #9 are only for RoutineControlType = requestRoutineResults and Supported Control / Routine Status = "finished".			

F.3.6 RID 0x0305- WheelSpeedSensorsTest

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.1
data #2	Maximum speed of the FL wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #3	Maximum speed of the FL wheel (low byte)	DEC	
data #4	Minimum speed of the FL wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #5	Minimum speed of the FL wheel (low byte)	DEC	
data #6	Maximum speed of the FR wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s

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Byte No.	Description	Data Type	value
data #7	Maximum speed of the FR wheel (low byte)	DEC	
data #8	Minimum speed of the FR wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #9	Minimum speed of the FR wheel (low byte)	DEC	
data #10	Maximum speed of the RL wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #11	Maximum speed of the RL wheel (low byte)	DEC	
data #12	Minimum speed of the RL wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #13	Minimum speed of the RL wheel (low byte)	DEC	
data #14	Maximum speed of the RR wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #15	Maximum speed of the RR wheel (low byte)	DEC	
data #16	Minimum speed of the RR wheel (high byte)	DEC	Resolution: 1/64/bit, offset: 0. range: -100~100m/s
data #17	Minimum speed of the RR wheel (low byte)	DEC	
NOTE: If any of the wheel speed sensor is invalid, the output value will be 0xFFFF. **NOTE:** These bytes from data #2 to data #17 are only for RoutineControlType = requestRoutineResults and Supported Control / Routine Status = "finished".			

F.3.7 RID 0x0402- InertialSensorCalibration (ESP only)

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	If routineControlType = startRoutine, then 01 = start successful 02 = already started 03 = not possible to start Else routineControlType = requestRoutineResults, then 01 = finished 02 = still running 03 = stopped or not start 04 = routine stopped abnormal
NOTE: This routineIdentifier 0x0402 cannot be stopped by routineControlType = Stop Control.			

F.3.8 RID 0x0403- Speed Limit Disable/Enable

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.7
NOTE: This request is Disable/Enable the speed Limit (10kph for Extended Mode).			

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Byte No.	Description	Data Type	value
NOTE: This routineIdentifier 0x0403 cannot be stopped by routineControlType = Stop Control.			

F.3.9 RID 0x0404- Disable Valve Relay Shutdown

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.7
NOTE: This routineIdentifier 0x0404 cannot be stopped by routineControlType = Stop Control.			

F.3.10 RID 0x0405- SAS Calibration (ESP only) (Optional for ESP)

Byte No.	Description	Data Type	value
data #1	Supported Control/ Routine Status	DEC	Same as the Comments in the table of F.3.7
NOTE: This routineIdentifier 0x0405 cannot be stopped by routineControlType = Stop Control.			

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Annex G (informative) Examples for addressAndLengthFormatIdentifier parameter values

The detailed description about this Annex can be found in Annex G of the document *CN200 General Diagnostic Requirement Specification*.

It is possible for ABS supplier to add the additional definitions in Annex G for CN200 vehicle, if necessary, it is ABS supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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